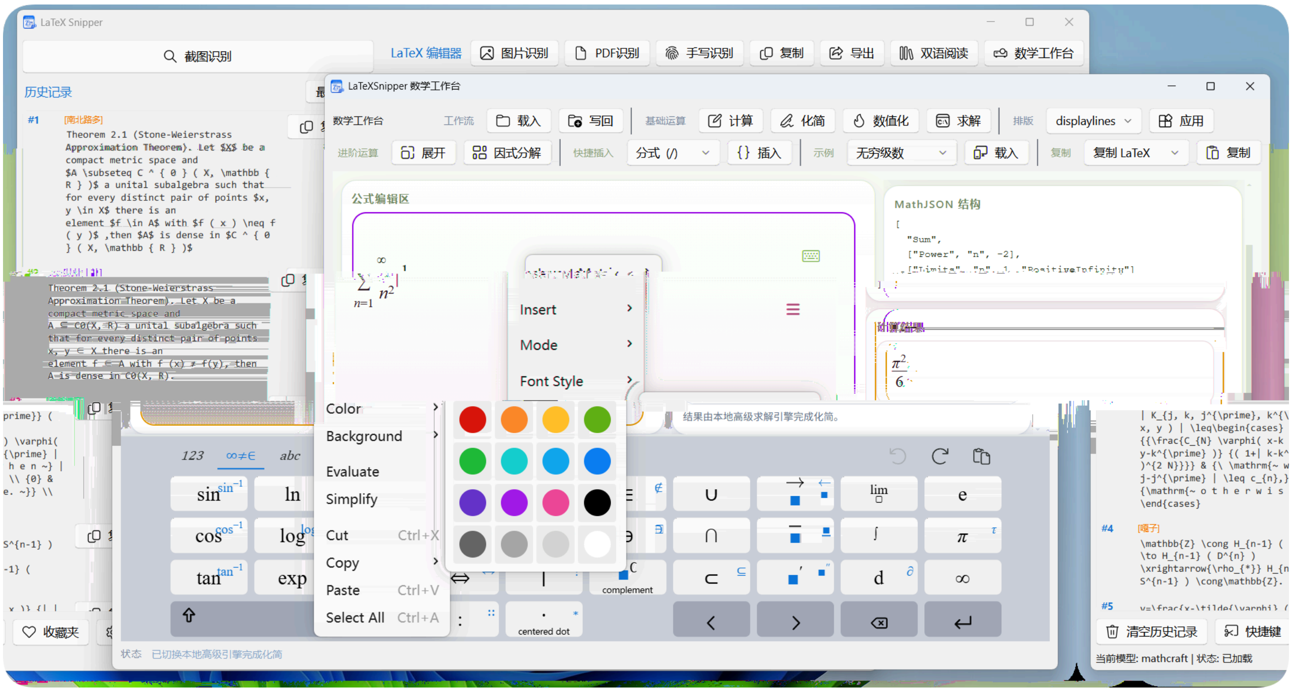
v2.3.2_Final_Stable |



-
- MathCraft ONNX
-
-
-
- PDF
- Typst
-
- Linux / macOS
-
-
- Bug

- MathCraft OCR
-

LaTeXSnipper “ ” ——
Python OCR

⚠️
“ ”
Python apt/brew

- Windows
- Linux `python3 -m venv` Debian/Ubuntu `python3-venv`
- macOS Python 3.10+ Homebrew Python python.org

1.
2. LaTeX
3. LaTeX
4. “ ”
5.
6. 30+ Pandoc

- `%USERPROFILE%\.latexsnipper\logs\` `%LOCALAPPDATA%\LaTeXSnipper\logs\`
- `~/.latexsnipper/logs/`
- `~/.latexsnipper/logs/`
- `crash-native.log`

✔️
Bug logs

---

warmup() warmup
python -m mathcraft_ocr models check

" model xxx downloading "

- GitHub Releases
- /
- urllib

- GitHub
- HTTP_PROXY / HTTPS_PROXY
-
- MathCraft
  - Windows `%APPDATA%\MathCraft\models\`
  - Linux/macOS `~/.mathcraft/models/`

`python -m mathcraft_ocr models check` `python -m mathcraft_ocr models download`

`failed to import onnxruntime` `missing get_available_providers`

- `onnxruntime`
- pip
- CPU GPU

- `pip uninstall onnxruntime onnxruntime-gpu`
- GPU `pip install onnxruntime-gpu`
- CPU `pip install onnxruntime`

warmup CUDA `cuda wasn't able to be loaded` `cublas`

- `onnxruntime-gpu` CUDA / cuDNN
- CUDA onnxruntime-gpu
-
- GPU

- MATHCRAFT_FORCE_ORT_CPU=1 CPU
- CUDA Toolkit cuDNN onnxruntime-gpu onnxruntime
-

warmup invalid protobuf failed to load model sha256 mismatch

- Windows %APPDATA%\MathCraft\models\<model_id>\ Linux/macOS ~/.mathcraft/models/<model_id>/
-
- python -m mathcraft_ocr models check

warmup CUDA out of memory
GPU

- GPU
- MATHCRAFT_FORCE_ORT_CPU=1 CPU

“ ” “ ”

✗

_validate_config() provider mineru model_name base_url model_name

Base URL “ ”

OCR 60

vLLM / Ollama / ONNX / 7B+ 60

- curl Ollama API
- 120-180
- Ollama `ollama run <model_name>`

“ 127.0.0.1:11434 ”

- `http://localhost:11434/v1` Ollama `/v1`
- `https://` `http://` Ollama HTTPS
- Ollama
- Ollama `ollama serve`

✔

`http://127.0.0.1:11434/api/tags` JSON

“ : xxx ”

- Ollama `ollama list`
- OpenAI-compatible `/v1/models`

401 “ ”

DeepSeek OpenAI API Key Ollama API Key

⚠

API Key LaTeXSnipper API Key
Base URL

404 409
MinerU `/health` `/file_parse`

- MinerU `http://127.0.0.1:8000/health`
- “ ” “ ”
- MinerU 409 MinerU

`/v1/models` 404
Ollama `/api/tags` `/v1/models`
“ OpenAI-compatible ” “ Ollama ”

- Ollama →
- / DeepSeek / OpenAI / Groq →
- MinerU →
- MathCraft ONNX →

LaTeX
“ LaTeX ”

Python

- Python Python 3.11
- Python 3.10+
- Python 3.11 Windows

```
# Python 3.11 创建虚拟环境
python3.11 -m venv .venv

# Windows 激活
.\.venv\Scripts\activate

# Linux/macOS 激活
source .venv/bin/activate

# 安装依赖
pip install -r requirements.txt
```

GitHub Release exe `git clone` Windows Python 3.11 Linux/macOS PyInstaller Python

- Release
- `requirements.txt` `requirements-build.txt`

Linux/macOS `src/deps/python311` MathCraft Pandoc Python

```
~/.latexsnipper/deps/python311
```

Linux/macOS `/usr/lib/latexsnipper` `.app` `~/.latexsnipper/deps/python311` venv `pyvenv.cfg` CI

Linux `.deb` `python3` `python3-venv` macOS `.deb` `python3`

```
# Homebrew
brew install python
```

```
# 或使用 python.org 官方 macOS 安装包
https://www.python.org/downloads/macos/
```

```
pip install -r requirements.txt
```

- pywin32 wheel
- PyQt6
- PyPI
- / PyPI

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple -r requirements.txt
```

- pywin32 pip PyPI wheel

C:\Program Files\

%APPDATA% ~/.latexsnipper/

UAC

-
- %LOCALAPPDATA%

Windows %APPDATA% ASCII C ONNX

MATHCRAFT_HOME ASCII C:\mathcraft_data

Ollama / MinerU LaTeXSnipper
Windows Defender

- Python / Ollama
-
- Ollama `0.0.0.0` `127.0.0.1`

`api.github.com`

-
- GitHub Releases
- `%USERPROFILE%/.latexsnipper/logs/`

exe
PyInstaller exe

-
- SignPath

PDF

- DPI < 100 > 200
- PDF
-

- PDF 140-170 DPI
- 200-300 DPI
- “ ”

LaTeX Markdown
MinerU

4K
Base64 33%

-
-
-

Typst LaTeX LaTeXSnipper Typst
→ → → Typst Pandoc

LaTeXSnipper

- —
- — MathJax LaTeX
- — MathJax
- — LaTeX SVG
- — xelatex SVG
- — Typst CLI SVG

Typst
→ pypandoc → → Typst CLI → →

⚠️
Typst Typst
MathJax Typst

LaTeX ↔ Typst LaTeXSnipper

- LaTeX → Typst Typst → LaTeX
-
-
-
-

LaTeX → Typst pypandoc Typst → LaTeX
——

" Typst " LaTeX/Typst

- Typst CLI PATH
- Typst
- Typst `--format svg`
- pypandoc LaTeX Typst

- Typst

- https://github.com/typst/typst

```
# Windows (scoop / winget)
winget install Typst.Typst

# macOS
brew install typst

# Linux
cargo install typst-cli
```

- → → " Typst " → " " Typst CLI → " "
- LaTeX Typst

```
pip install pypandoc
```

Pandoc https://pandoc.org

- Typst CLI `typst --version` ≥ 0.11

" "

- Typst
-
- Typst

- `typst --version`
- Windows `C:\Users\<用户名>\.cargo\bin\typst.exe` cargo `C:\Program Files\Typst\typst.exe` winget
- macOS/Linux `/usr/local/bin/typst` `~/.cargo/bin/typst`
- PATH typst

Typst

- pypandoc LaTeX → Typst LaTeX
- Typst LaTeX `\mathbb{R}` vs `RR`
- Typst LaTeX

- " " " MathJax " " LaTeX + pdflatex/xelatex "
- Typst Typst
- `[Typst] Pandoc conversion` `[ERROR] Typst compile failed`

> pypandoc latex→typst LaTeX
>
> - `\newcommand`
> - AMSmath `\substack` `\xrightarrow`

- /
- LaTeX

MathJax LaTeX

15

Typst 15

- MathJax
- /CPU
- Typst

Pandoc `.typ` Typst CLI

- Pandoc typst Typst
- Pandoc LaTeX
- Typst Pandoc

- Pandoc ≥ 3.0 `pandoc --version`
- Typst ≥ 0.11
- `.typ`
- LaTeX

LaTeXSnipper MathCraft OCR Markdown `typst_formulas` `True` LaTeX Typst

- `pypandoc` `pip install pypandoc`
- Pandoc
- pypandoc LaTeX

✔

Python

```
python -c "from core.mathcraft_document_engine import convert_latex_to_typst;
print(convert_latex_to_typst(r'\\int_{0}^{\\infty} e^{-x} dx'))"
```

-
- Wayland
-

Word / EPUB / Typst
Pandoc Pandoc

- “ PANDOC ”
- Pandoc https://pandoc.org PATH
- LaTeX / Markdown / MathML / HTML Pandoc

- OCR
-
-

-
- “ ”
-

-
- “ ”

> - 700ms > 3 300ms ——
> - 250ms ≤ 2 ——
> - 350ms ——

- \+ OCR
- UnsharpMask
- QImage → PIL PNG
- < 120px 2x

- *
- Compute Engine

- `2x` `2*x`
- SymPy/mpmath
-

[LaTeX]

- Typst Typst
- LaTeX
- v2.3.2 LaTeX

Typst
`[LaTeX]` LaTeX Typst

-
- “ ” “ ” “ ”

`.docx` `.epub`
Pandoc UTF-8

- Pandoc ≥ 3.0
- LaTeX PDF `ctex`
- Word Word

Wayland
Wayland Qt

- grim + slurp wlroots gnome-screenshot GNOME
-
-

> Python grim maim gnome-screenshot
> —— Qt

EGL not available No physical devices GLOzone not found Failed to get system egl display
MathCraft GPU Qt WebEngine / Chromium Wayland WSL DRI render EGL/GPU
LaTeXSnipper Linux X11 + GPU + /dev/dri/renderD*

```
# 强制启用 Linux 图形兜底
LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS=1 latexsnipper

# 禁用 Linux 图形兜底，尝试原生 Qt/GPU 路径
LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS=1 latexsnipper
```

XWayland/XCB/QtWebEngine Debian/Ubuntu

```
sudo apt install xwayland libxcb-cursor0 libxcb-icccm4 libxcb-image0 \
  libxcb-keysyms1 libxcb-render-util0 libxcb-xinerama0 libxkbcommon-x11-0
```

macOS “ ” “ ”
Apple notarization

- → → “ ”
- xattr -cr /Applications/LaTeXSnipper.app

- → → LaTeXSnipper

-

pip pip / setuptools / wheel
macOS Python Python 3.10+
~/.latexsnipper/deps/python311 Python venv / ensurepip / pip

- Homebrew Python brew install python
- python.org macOS Python
- LaTeXSnipper

```
配置文件 & 日志:
  Windows:  %USERPROFILE%\.latexsnipper\
  Linux:    ~/.latexsnipper/
  macOS:    ~/.latexsnipper/

Linux/macOS Python 依赖环境:
  Linux:    ~/.latexsnipper/deps/python311
  macOS:    ~/.latexsnipper/deps/python311

模型缓存 (MathCraft ONNX):
  Windows:  %APPDATA%\MathCraft\models\
  Linux:    ~/.mathcraft/models/
  macOS:    ~/.mathcraft/models/
```

LaTeXSnipper

- LaTeXSnipper
- Python

- Explorer CMD
- open Finder Terminal.app
- xdg-open x-terminal-emulator

GUI

Python PATH Python

- `where python` Windows / `which python` Linux/macOS Python
-
-

hook

LaTeXSnipper

✗

= =

Bug

%USERPROFILE%\.latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnipper\logs\

stderr

crash-native.log

exe Python

- https://github.com/SakuraMathcraft/LaTeXSnipper/issues
- https://github.com/SakuraMathcraft/LaTeXSnipper/discussions

---

README

✔

- python -m venv .venv → pip install -r requirements.txt → python src/main.py
- python3 -m venv .venv → pip install -r requirements-linux.txt → python src/main.py
- python3 -m venv .venv → pip install -r requirements-macos.txt → python src/main.py
- requirements-linux.txt requirements-macos.txt -r requirements.txt
  pynput

python311/ src/deps/python311/ Python

- python311/ Windows Python
  ruff/pyright/pytest
- src/deps/python311/ IDE Actions requirements*.txt
  ruff pytest pyright
- Linux/macOS src/deps/python311 ~/.latexsnipper/
  deps/python311

```
py -3.11 -m venv src\deps\python311
.\src\deps\python311\python.exe -m pip install --upgrade pip wheel setuptools
.\src\deps\python311\python.exe -m pip install -r requirements.txt
.\src\deps\python311\python.exe -m pip install -r requirements-build.txt
.\src\deps\python311\python.exe -m pip install ruff pytest pyright
```

- python311/ — Windows
- src/deps/python311/ —
- ~/.latexsnipper/deps/python311 — Linux/macOS

docs/developer_code_standards.md

```
.\src\deps\python311\python.exe -m ruff check .
.\src\deps\python311\python.exe -m pytest test
.\src\deps\python311\python.exe -m pyright
.\src\deps\python311\python.exe -m compileall -q -x "src[\\/]+deps" src mathcraft_ocr test
```

⚠

ruff pytest pyright requirements*.txt

```
.\src\deps\python311\python.exe -m pip install ruff pytest pyright
```

`pyrightconfig.json`

`src/deps/python311` `python311/` Windows
pyright

> `src/deps/python311` `python311`

MathCraft OCR LaTeXSnipper ONNX OCR

- GitHub https://github.com/SakuraMathcraft/MathCraft-Models
- PyPI https://pypi.org/project/mathcraft-ocr/

-
- ONNX Runtime CPU CUDA GPU
- PDF
-

```
pip install mathcraft-ocr
```

LaTeXSnipper

```
# 检查模型缓存状态
python -m mathcraft_ocr models check

# 手动下载模型
python -m mathcraft_ocr models download

# 识别图片中的公式
python -m mathcraft_ocr run --image formula.png
```

" MathCraft ONNX "

CPU GPU

`mathcraft-ocr` PyPI `0.1.8` MathCraft Models `v1.0.0`

- `mathcraft-formula-det` — Formula Detection
- `mathcraft-formula-rec` — LaTeX Formula Recognition
- `mathcraft-text-det` — Text Detection
- `mathcraft-text-rec` — Text Recognition

MathCraft → → + →
→

- **`formula`** — → LaTeX
- **`text`** — OCR →
- **`mixed`** — + → Markdown

PROOF. (1) If $A \cup B = \emptyset$, then the theorem follows immediately since both A and B are the empty set. Otherwise, we must show that $(A \cup B)' \subset A' \cap B'$ and $(A \cup B)' \supset A' \cap B'$. Let $x \in (A \cup B)'$. Then $x \notin A \cup B$. So x is neither in A nor in B, by the definition of the union of sets. By the definition of the complement, $x \in A'$ and $x \in B'$. Therefore, $x \in A' \cap B'$ and we have $(A \cup B)' \subset A' \cap B'$.

To show the reverse inclusion, suppose that $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$, and so $x \notin A$ and $x \notin B$. Thus $x \notin A \cup B$ and so $x \in (A \cup B)'$. Hence, $(A \cup B)' \supset A' \cap B'$ and so $(A \cup B)' = A' \cap B'$.

The proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset.$$

To see that this is true, observe that

$$\begin{aligned}(A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset.\end{aligned}$$

Cartesian Products and Mappings

Given sets A and B, we can define a new set $A \times B$, called the ***Cartesian product*** of A and B, as a set of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \emptyset$, then $A \times B$ is the set

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

and

$$A \times C = \emptyset.$$

We define the ***Cartesian product of n sets*** to be

$$A_1 \times \cdots \times A_n = \{(a_1, \ldots, a_n) : a_i \in A_i \text{ for } i = 1, \ldots, n\}.$$

If $A = A_1 = A_2 = \cdots = A_n$, we often write A^n for $A \times \cdots \times A$ (where A would be written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ are called ***relations***. We will define a ***mapping*** or ***function*** $f \subset A \times B$ from a set A to a set B to be the special type of relation where each element $a \in A$ has a unique element $b \in B$ such that $(a, b) \in f$. Another way of saying this is that for every element in A, f assigns a unique element in B. We usually write $f : A \to B$ or $A \xrightarrow{f} B$. Instead of writing down ordered pairs $(a, b) \in A \times B$, we write $f(a) = b$ or $f : a \mapsto b$. The [redacted] A is called the ***domain*** of f and [redacted] set [redacted]

$$f(A) = \{f(a) : a \in A\} \subset B$$

[redacted] is called the ***range*** or ***image*** of f. We can think of the elements in the function's domain as input values and the elements in the function's range as output values.

[redacted]

where

$$
\begin{aligned}
&F_{11}(u_n, v_n, \delta) \\
&= -\frac{(1+b)x^*\delta}{2a} - \frac{a\sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\
&\quad - \frac{a\cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\
&F_{12}(u_n, v_n, \delta) \\
&= -\frac{\left(-x^*b^2 + 2a\sin x^* - 2x^*b - x^*\right)\delta}{2a} \\
&\quad + \frac{a(b-1)\sin x^* u_n^2}{2} + \frac{(1+b)^2 x^*\delta^2}{4a^2} - \cos x^*\delta u_n \\
&\quad + \frac{a(b-1)\cos x^* u_n^3}{6} + \frac{(1+b)^2 x^*\delta^3}{8a^3} + \frac{\sin x^*\delta u_n^2}{2} \\
&\quad + O(\|(u_n, v_n, \delta)\|^4).
\end{aligned}
$$

Consider the following system

$$
\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \widehat{F}_{11}(u_n, v_n, \delta_n) \\ \widehat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \tag{15}
$$

where

$$
a_{11} = -1 + a\cos x^*, \quad a_{12} = 1,
$$

[redacted]

then the system (15) becomes

$$
\begin{aligned}
\begin{pmatrix} \tilde{x}_{n+1} \\ \tilde{y}_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a\cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} \\
&\quad + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix},
\end{aligned} \tag{16}
$$

where

$$
\begin{aligned}
&\tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\
&= -\frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2\cos x^*(a - 2b - 2)} \\
&\quad - \frac{a\sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a\sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a\cos x^* - b - 1)} \\
&\quad - \frac{\tilde{\delta}_n^2}{8(a\cos x^* - b - 1)^2 a^2}\Big(\big(2x^*(b+1)^3 + a^3\sin^3 x^* \\
&\quad - 4ax^*(b+1)^2\cos x^* + 2a^2 x^*(b+1)\cos^2 x^*\big)\Big) \\
&\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a\cos x^* - b - 1)}
\end{aligned}
$$

[redacted]

第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

作者联系方式：QQ:937821418

本讲义将自动生成和 cctalk 用户绑定的 ID，如果你的 ID 多次出现在公开环境，我们会追究法律责任.

Problem Books in Mathematics

Ovidiu Furdui

Limits, Series, and Fractional Part Integrals

Problems in Mathematical Analysis

Springer

block_layout_regression_v4 CUDA

- 10
- 495
- 21,417
- Markdown 304
-
- 1.33
- 18.53

MathCraft OCR

- ONNX Runtime PyTorch
- manifest
- /
- OCR
-

MATHCRAFT_FORCE_ORT_CPU MATHCRAFT_HOME

```
Windows (永久生效):
  1. Win+R → 输入 sysdm.cpl → 高级 → 环境变量
  2. 在"用户变量"中新建, 变量名填入上面给出的名称, 值填入 1 或路径

Windows (仅当前终端):
  在 PowerShell 中执行: $env:MATHCRAFT_FORCE_ORT_CPU = "1"

Linux / macOS (永久生效):
  在 ~/.bashrc 或 ~/.zshrc 末尾添加:
  export MATHCRAFT_FORCE_ORT_CPU=1
  然后执行 source ~/.bashrc

Linux / macOS (仅当前终端):
  export MATHCRAFT_FORCE_ORT_CPU=1
```

- MATHCRAFT_FORCE_ORT_CPU — 1 CPU GPU
- MATHCRAFT_HOME — MathCraft
- HTTP_PROXY / HTTPS_PROXY —
- MATHCRAFT_LOG_LEVEL — DEBUG / INFO / WARNING / ERROR
- LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS — Linux Qt/WebEngine
- LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS — Linux Qt/GPU
- LATEXSNIPPER_SHOW_CONSOLE — Windows